(9) $\int \frac{\sec x}{\sqrt{\cos 2x}} dx$ is

(1) $\tan^{-1}(\sin x)+c$ (2) $2\sin^{-1}(\tan x)+c$ (3) $\tan^{-1}(\cos x)+c$ (4) $\sin^{-1}(\tan x)+c$

(10) $\int \tan^{-1}\sqrt{\frac{1-\cos 2x}{1+\cos 2x}}\, dx$ is

(1) x^2+c (2) $2x^2+c$ (3) $\frac{x^2}{2}+c$ (4) $-\frac{x^2}{2}+c$

(11) $\int 2^{3x+5} dx$ is

(1) $\frac{3(2^{3x+5})}{\log 2}+c$ (2) $\frac{2^{3x+5}}{2\log(3x+5)}+c$ (3) $\frac{2^{3x+5}}{2\log 3}+c$ (4) $\frac{2^{3x+5}}{3\log 2}+c$

(12) $\int \frac{\sin^8 x-\cos^8 x}{1-2\sin^2 x\cos^2 x}\, dx$ is

(1) $\frac{1}{2}\sin 2x+c$ (2) $-\frac{1}{2}\sin 2x+c$ (3) $\frac{1}{2}\cos 2x+c$ (4) $-\frac{1}{2}\cos 2x+c$

(13) $\int \frac{e^x(x^2\tan^{-1}x+\tan^{-1}x+1)}{x^2+1}\, dx$ is

(1) $e^x\tan^{-1}(x+1)+c$ (2) $\tan^{-1}(e^x)+c$ (3) $e^x\frac{(\tan^{-1}x)^2}{2}+c$ (4) $e^x\tan^{-1}x+c$

(14) $\int \frac{x^2+\cos^2 x}{x^2+1}\operatorname{cosec}^2 x dx$ is

(1) $\cot x+\sin^{-1}x+c$ (2) $-\cot x+\tan^{-1}x+c$

(3) $-\tan x+\cot^{-1}x+c$ (4) $-\cot x-\tan^{-1}x+c$

(15) $\int x^2\cos x\, dx$ is

(1) $x^2\sin x+2x\cos x-2\sin x+c$ (2) $x^2\sin x-2x\cos x-2\sin x+c$

(3) $-x^2\sin x+2x\cos x+2\sin x+c$ (4) $-x^2\sin x-2x\cos x+2\sin x+c$

(16) $\int \sqrt{\frac{1-x}{1+x}} dx$ is

(1) $\sqrt{1-x^2}+\sin^{-1}x+c$ (2) $\sin^{-1}x-\sqrt{1-x^2}+c$

(3) $\log\left|x+\sqrt{1-x^2}\right|-\sqrt{1-x^2}+c$ (4) $\sqrt{1-x^2}+\log\left|x+\sqrt{1-x^2}\right|+c$

(17) $\int \frac{dx}{e^x-1}$ is

(1) $\log|e^x|-\log|e^x-1|+c$ (2) $\log|e^x|+\log|e^x-1|+c$

(3) $\log|e^x-1|-\log|e^x|+c$ (4) $\log|e^x+1|-\log|e^x|+c$

11_11th-Maths-Vol2-Unit11.indd 226 16-12-2022 11:08:18